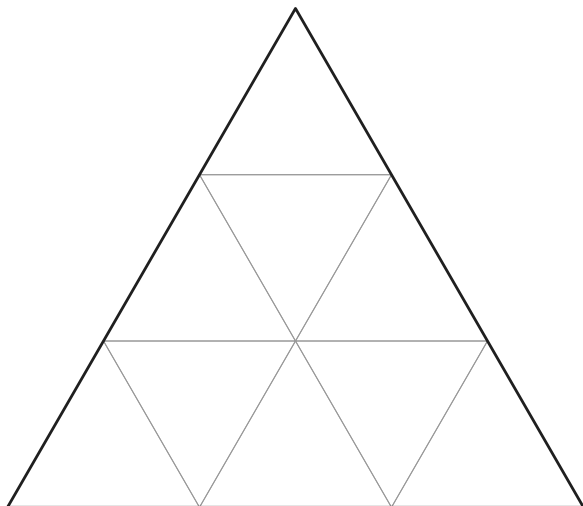


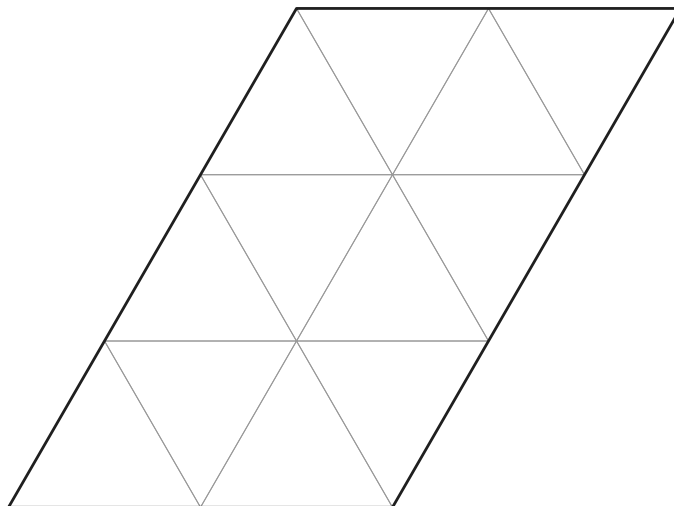
The Great Board Mystery

Problem 1: Try to fill each board using only small blue rhombi. Each blue covers two whole triangle spaces. Keep pieces inside the board without overlaps. Remember: tiling isn't just about filling space, it's about seeing hidden patterns!

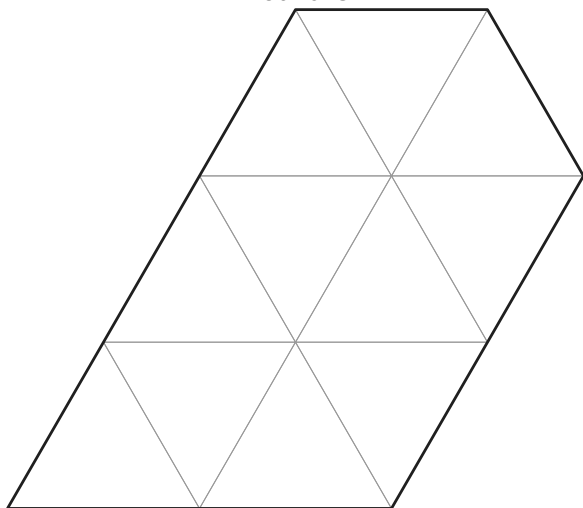
Board A



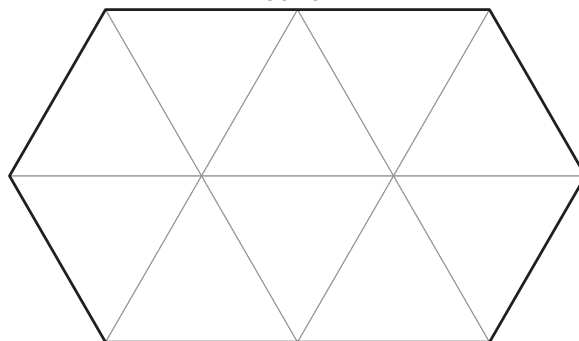
Board B



Board C



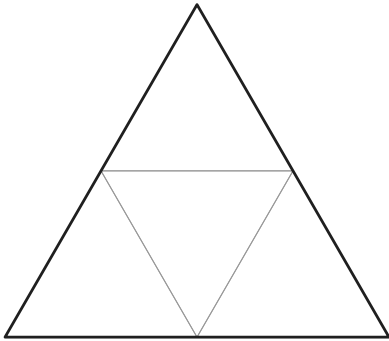
Board D



Which boards can you fill and which ones did you get stuck on? What is stopping you from completely filling some of the shapes?

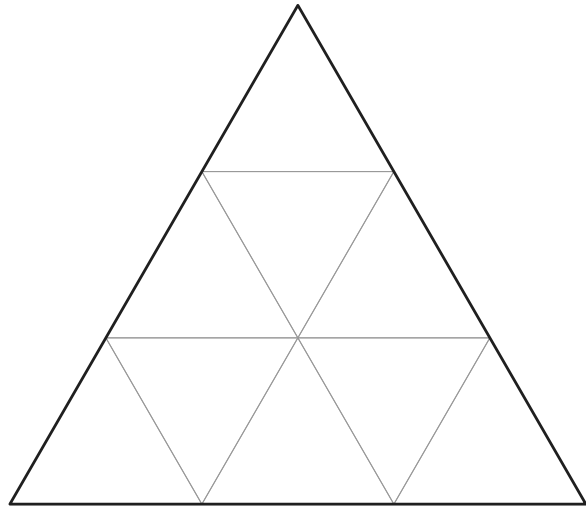
Problem 2: Use only small blue rhombi on these triangles and the triangle on the next page. Fit as many as you can, without overlaps or pieces outside. Record how many blues fit and how many triangle spaces stay empty. Hint: Color the triangles that point up and the triangles that point down, then count each kind.

2 small edges on each side



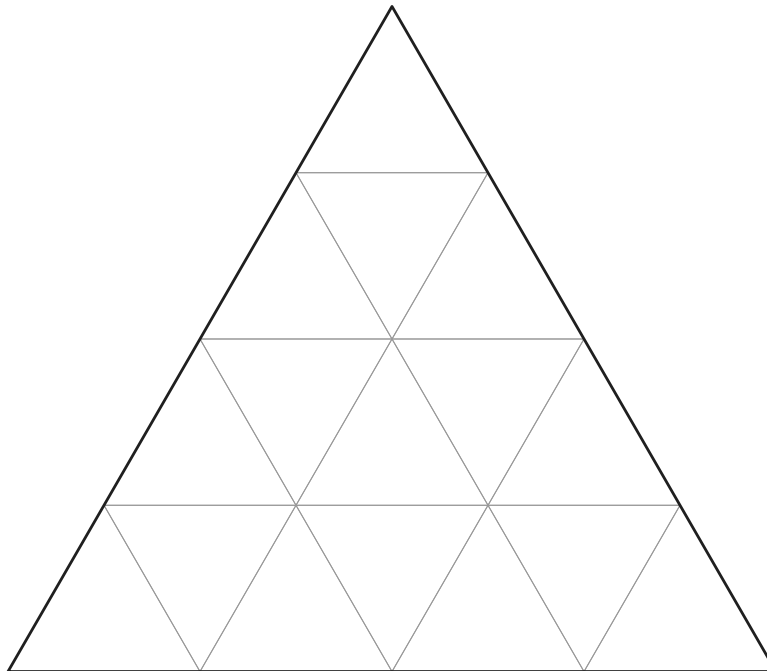
Blues: _____ Empty spaces: _____

3 small edges on each side



Blues: _____ Empty spaces: _____

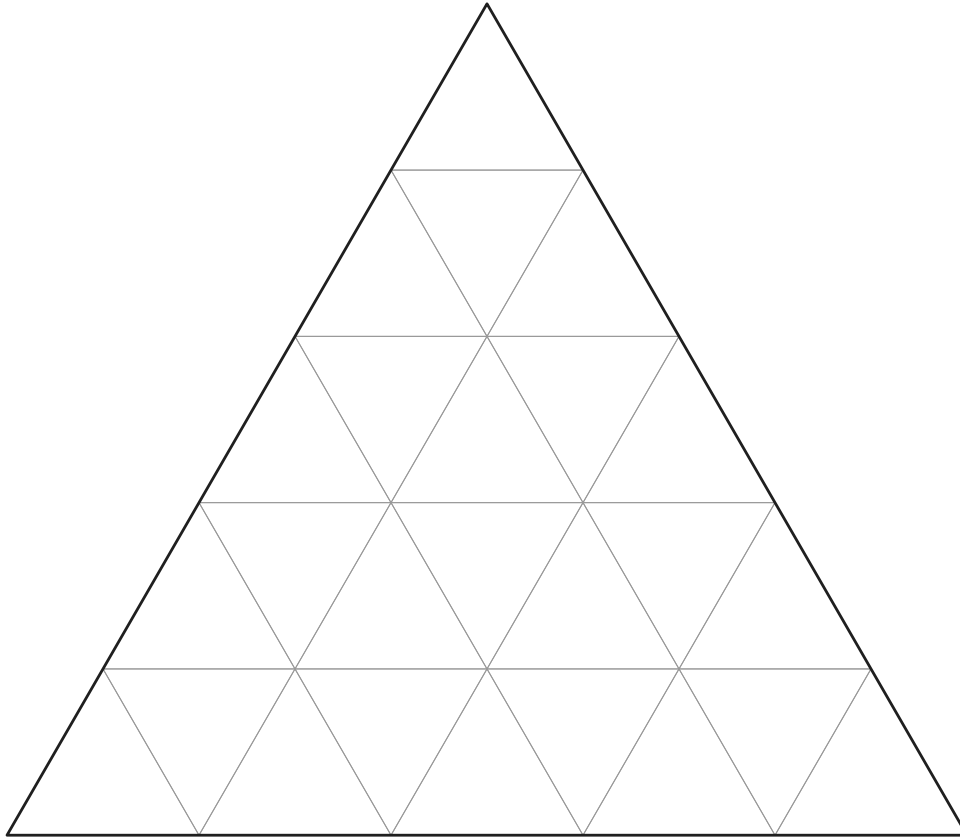
4 small edges on each side



Blues: _____ Empty spaces: _____

Problem 2: Fit as many small blue rhombi as you can in this triangle. Keep pieces inside the triangle without overlaps. Record your best packing.

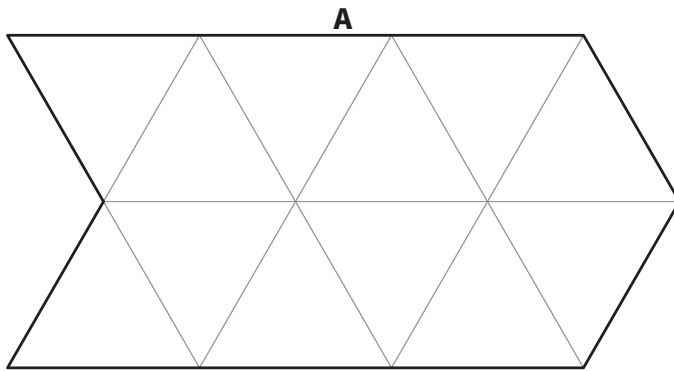
5 small edges on each side



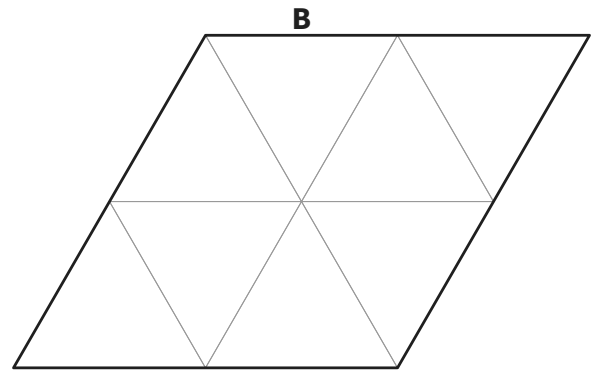
Blues: _____ Empty spaces: _____

Compare your four best packings. What do you notice about the number of empty spaces? Explain why you cannot leave fewer.

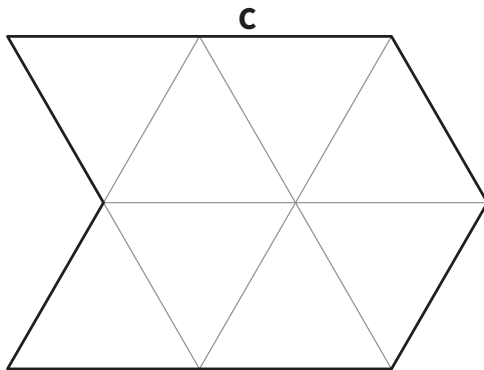
Problem 3: On each board, first use only small blue rhombi. Then clear it and use only purple chevrons. Cover whole triangle spaces without overlaps or pieces outside. Leave as few spaces empty as you can. Record your best results.



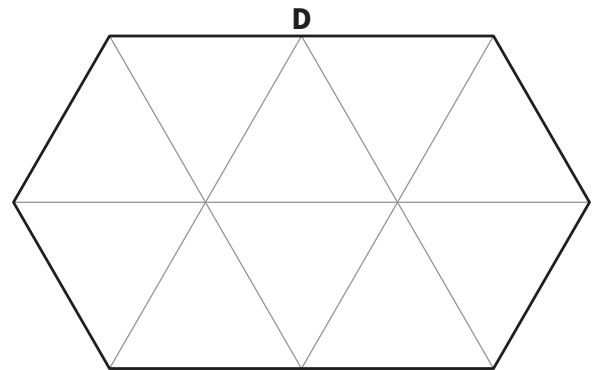
Fewest empty triangle spaces:
Blues: ____ Purples: ____



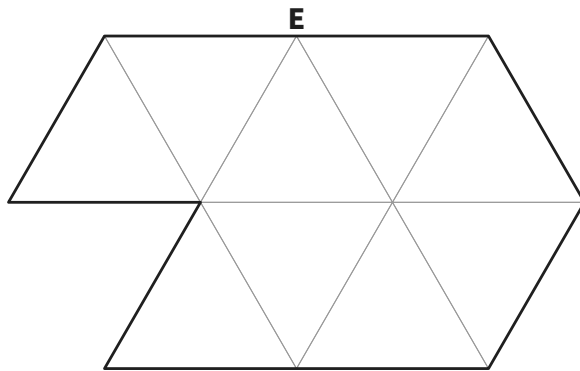
Fewest empty triangle spaces:
Blues: ____ Purples: ____



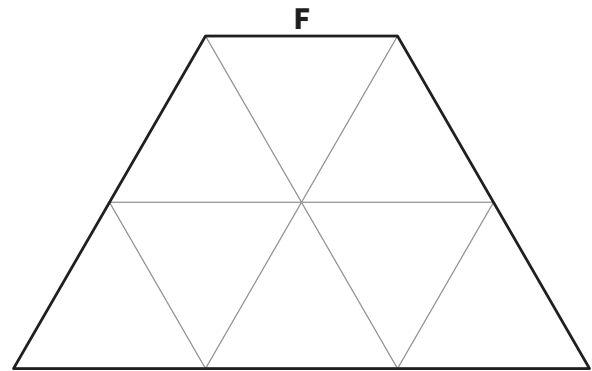
Fewest empty triangle spaces:
Blues: ____ Purples: ____



Fewest empty triangle spaces:
Blues: ____ Purples: ____



Fewest empty triangle spaces:
Blues: ____ Purples: ____



Fewest empty triangle spaces:
Blues: ____ Purples: ____

Can you cover one purple exactly with blues? How does this explain which pieces can leave fewer gaps?

Fun fact: Mathematicians call this kind of argument a coloring proof!

